SUMMER INTERNSHIP PROJECT

ELECTRICITY BOARD BILLING SYSTEM

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Academic Year 2025–2026

PROJECT OVERVIEW

- The Electricity Board Billing System is designed to organize electricity consumer and billing activities in one application.
- The project focuses on consumer details, electricity connections, meter readings, bill generation, payments and receipts.
- It also includes complaint handling and notifications to support communication between consumers and the electricity board.
- The goal is to make billing information easier to maintain, access and track.

PROBLEM STATEMENT

- Managing consumer records, meter readings and payment details separately can make billing work difficult to track.
- Manual entry may lead to repeated work and errors when calculating consumption or preparing bills.
- It can also be difficult to monitor due dates, unpaid bills and the status of complaints.
- A centralized system can keep related information organized and make common tasks easier to follow.

PROJECT OBJECTIVES

- Maintain consumer profiles and electricity connection information.
- Record meter details and meter readings, including units consumed and reading dates.
- Generate bills containing bill dates, due dates and total amounts.
- Record payments and generate or view payment receipts.
- Allow consumers to register complaints and follow their status.
- Support notifications and administrator/staff activities.

SYSTEM USERS AND ROLES

- Consumer: views bills, makes payments, receives notifications and registers complaints.
- Administrator: manages consumer information and generates bills.
- Staff: records meter readings and helps resolve complaints.
- The system stores the related consumer, connection, meter, reading, bill, payment, receipt and complaint information.
- Access should be limited according to each user's role.

MAJOR SYSTEM MODULES

- Consumer and Connection: stores consumer contact details and connection type/status.
- Meter and Meter Reading: stores meter information and records readings and units consumed.
- Billing: uses recorded consumption to calculate and generate a bill with dates and amount.
- Payment and Receipt: records payment amount, method and date, then supports receipt generation.
- Complaint and Notification: records complaints, tracks their status and sends messages.
- Administrator and Staff: support bill management, reading entry and complaint resolution.

BASIC BILLING WORKFLOW

- Step 1: Register or locate the consumer and confirm the connection details.
- Step 2: Record the meter reading and reading date.
- Step 3: Determine units consumed from the current and previous readings.
- Step 4: Calculate the bill amount according to the applicable tariff rules.
- Step 5: Generate the bill with its bill date and due date.
- Step 6: Record the consumer's payment and issue a receipt.
- Step 7: Keep the bill and payment status updated for future reference.

DATA AND RECORDS

- Consumer: consumer ID, name, address, mobile number and email.
- Connection: connection ID, connection type and status.
- Meter Reading: reading ID, units consumed and reading date; meter records identify meter details.
- Bill: bill ID, bill date, due date and total amount.
- Payment and Receipt: payment ID, amount, method, date, receipt number and issue date.
- Complaint and Notification: complaint description/status and message/send date.
- These records should be linked consistently so a bill and payment can be traced to the correct consumer.

EXPECTED BENEFITS AND TESTING

- Organized consumer, meter, bill and payment records.
- Easier tracking of consumption, due dates, payment history and receipts.
- A clear process for registering and resolving complaints.
- Less repetitive manual work when the system is implemented and used correctly.
- Important checks include valid login, correct consumer details, reading entry, bill calculation, payment recording, receipt generation and complaint submission.
- Testing should also cover missing fields, invalid input and attempts to access functions without the correct role.

CONCLUSION

ELECTRICITY BOARD BILLING SYSTEM

The project brings consumer records, meter readings, bill generation, payments, receipts, complaints and notifications into one organized workflow. It aims to make billing records easier to maintain and track.

THANK YOU

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